

Kemal Adlığ

AI-Native Product Engineer · Full-Stack · Mobile · Cloud

Istanbul, Türkiye (Remote-first) | kemaladligdev@gmail.com | +90 553 188 21 50
github.com/kemaladlig | linkedin.com/in/kemaladlig

SUMMARY

Product engineer who ships production software alone, by orchestrating AI agent workflows. Recent work spans a published cross-platform mobile app, an AI document-processing SaaS for accounting firms, an MCP server for semantic code search, and a dozen smaller products across web, mobile, desktop, and real-time multiplayer. Several sit behind CI running over **450 automated tests**. Most days that means writing the specs and context maps agents build from, wiring MCP tools into the editor, and running evaluation harnesses against LLM features before they ship.

TECHNICAL SKILLS

AI & Agents	Multi-agent orchestration (Claude Code, Codex, OpenCode) · Context engineering (AGENTS.md / PROJECT_MAP.md specs) · MCP server development · Evaluation harnesses (recall@k, MRR) · RAG & semantic search (Ollama, EmbeddingGemma) · LLM integration (Gemini Vision, OpenAI)
Languages	TypeScript, JavaScript, Python, Kotlin, Dart, SQL, HTML/CSS
Frontend & Mobile	React 19, React Native (Expo), Next.js, Vite, Tailwind CSS v4, Zustand, Kotlin / Jetpack Compose, Capacitor, PWA
Desktop & Systems	PyQt6, ONNX Runtime (on-device OCR), NumPy, Win32 extended window styles, multithreaded workers
Backend & Data	Node.js, Bun, Supabase (Postgres, Row Level Security, Edge Functions), Firebase (Firestore, Cloud Functions), SQLite / Drizzle ORM, REST, WebSocket, WebRTC
Cloud & DevOps	Docker, Kubernetes (GKE / Huawei CCE), GitHub Actions, Nginx Ingress, Terraform, Vercel
Testing	Vitest, Jest, Playwright, Node test runner, CI quality gates

EXPERIENCE

Independent Product Engineer & Consultant Jan 2024 – Present
Self-employed · Remote

- Designed, built, and still operate several production products alone: a published cross-platform mobile app on the App Store and Google Play, an AI document-processing SaaS for accounting firms, and a Chrome extension. The stack spans React Native, Next.js, Supabase, and Firebase.
- Run parallel coding agents from versioned spec files (AGENTS.md and PROJECT_MAP.md context maps), and review every diff before it merges.
- Built **code-rag**, a Model Context Protocol server that gives any MCP client (Claude Code, Cursor, VS Code, Codex) local semantic code search. It has no npm dependencies and ships its own evaluation harness: **94% recall@5 and MRR 0.83** on a 16-question benchmark.
- Handle the whole release per product, from schema and authorization rules through test suites and CI to store review.

Software Engineer Intern Nov 2022 – Sep 2023
Huawei R&D Center · Istanbul, Türkiye

- Developed and optimized native Android apps in Kotlin, with attention to startup time, memory, and battery use.
- Integrated Huawei Mobile Services kits (Location, Push, Analytics, ML Kit) into production applications.
- Built Node.js and MongoDB services that processed **50,000+ rows** for an internal analytics tool, and designed and tested the REST APIs behind them with Postman.
- Worked in an Agile team, with Git-based code review.

Mobile Developer Intern Jul 2022 – Sep 2022
Birfen Elektrik Elektronik · Istanbul, Türkiye

- Built a mobile controller application bridging its UI to a robot's **ROS** backend over real-time protocols (TCP/IP, Bluetooth).

SELECTED PROJECTS

code-rag: MCP Server for Semantic Code Search github.com/kemaladlig/code-rag TypeScript / Node
Local MCP server so an agent can find code by meaning before it reads files. One install serves every project.

- No npm dependencies: Node's built-in `node:sqlite` with float32 blobs and brute-force cosine search, embeddings from EmbeddingGemma v1 via Ollama.

- Authored a **16-question evaluation harness** and used it to benchmark chunking and model choices, reaching **75% recall@1** and **94% recall@5** while cutting index time from about 20 minutes to 112 seconds.
- Derives the repo root from MCP roots or the working directory and keeps a self-ignored index per repo, so projects need no wiring.

Brutal Party: Real-Time Multiplayer Party Platform

Vanilla ES Modules · WebRTC

Four-player party mini-games in one lightweight app, with 15 game engines including Pong, Tanks, Archery, and Zone capture.

- Three networking modes from one codebase: single-device local play, LAN over a host WebSocket, and WebRTC peer-to-peer online, with Supabase used only for discovery and signaling.
- Star topology with separate control and world data channels, tuned to 8 Hz state sync and 30 Hz world broadcast with a jitter buffer, sequence and stale handling, and host-authoritative simulation.
- Built the CI gate in GitHub Actions: TypeScript check, four linters, **453 unit tests**, a 15-game device-independence quality gate (I1–I7 invariants), production build, and a Playwright engine matrix. It replaced a manual process where red tests piled up unnoticed.

Screen Translator: Real-Time In-Place Game Overlay github.com/kemaladlig/screen-translator

Python · PyQt6 · ONNX

Windows desktop tool that finds foreign-language text in full-screen games and paints each translation over the original pixels. The overlay is click-through, so the game stays playable underneath.

- On-device OCR through **RapidOCR** and ONNX Runtime returns pixel bounding boxes, so every translation lands on its own button or menu label instead of being collected in a side panel.
- Hardware click-through via Win32 extended window styles (`WS_EX_TRANSPARENT` | `WS_EX_LAYERED`), so the overlay never intercepts a click.
- A NumPy screen-diff engine (128×72 downsample, under 1 ms) skips OCR when the frame has not changed, and a SQLite cache answers repeated UI strings instantly. About 1,600 lines across a layered `core/ overlay/ ui/` split, with QThread keeping the PyQt6 interface responsive.

FişAktor: AI Document-Processing SaaS for Accounting Firms

Next.js · Supabase · Gemini

Turkish SaaS that reads receipts and invoices arriving through WhatsApp and the web, then exports balanced journal entries in the formats Luca and Zirve accept.

- Used **Gemini Vision** to pull date, vendor, tax number, and the 1% / 10% / 20% VAT split out of crumpled, angled, or badly lit photos.
- Falls back across models (`3.6-flash` → `3.5-flash` → `3.5-flash-lite`) so a quota limit never fails a user's upload.
- The output layer writes Excel and XML journal entries that import straight into the two dominant Turkish accounting suites, turning a manual minute per receipt into about two seconds.

Additional Production Work

- **VaultNote**: zero-server, end-to-end encrypted notes app (React 19, Capacitor). Data lives in the user's own Google Drive `appDataFolder`, so only ciphertext leaves the device. WebCrypto AES-256-GCM with HKDF and Argon2id hashing, covered by 66 unit tests and 17 end-to-end flows.
- **Gods of Rift**: deterministic strategy RPG on a Bun monorepo. Game logic sits in a dependency-free pure-TypeScript core, so the same seed always produces the same battle, and a PixiJS engine replays each simulation as a skippable animation.
- **ATA Akademi**: scheduling and billing platform for an arts academy (React 19, Supabase). Authorization is enforced by PostgreSQL Row Level Security, and a make-up-lesson algorithm picks the instructor's adjacent free slots, validated against declared working windows and a 24-hour cancellation rule.
- **KpssArena**: offline-first exam-prep app (Expo, Supabase). A SQLite and Drizzle store syncs to Postgres through `ccloudId` mapping, competitive modes run on an Edge Functions ghost opponent, and monetization is AdMob plus RevenueCat.
- Also shipped: **KubectlCommandTool** (a `kubect1` reference deployed to Google Kubernetes Engine with Docker, GitHub Actions CI/CD and Nginx Ingress TLS), **Rahmet Eli** (published Islamic companion app, Expo and Firebase, through App Store and Google Play review), **TraceX** (zero-backend crypto terminal on live Binance WebSocket and on-chain MVRV feeds), **EdgeBar** (Manifest V3 Chrome extension, no runtime dependencies), **Exam Timer** (PWA with YKS/KPSS/ALES presets), **Cloud Path** (cooperative Three.js game), **Civar**, **SyncCraft**, and **Auto-Motion**.

EDUCATION & CERTIFICATIONS

B.S. Computer Engineering, Yalova University

2020 – 2024

- Huawei Cloud **HCCDP** (Solution Architecture) and **HCCDA** (Cloud Native), 2025, through the Huawei Cloud DevOps Bootcamp.
- 3rd Place, Huawei Coding Marathon (2022), nationwide mobile-services category.
- Technical Author, Huawei Developers Medium Blog: "Lambda Expressions and Higher-Order Functions in Kotlin."